DSA_2024_PYQ

Group-A (Very Short Answer Type Question)

1.

a) Which type of linked list contains a pointer to the next as well as the previous node in the sequence?

b) The time complexity of heap sort in worst case is _____.

c) The running time complexity of a linear time algorithm is given as _____.

d) A linear list in which elements can be added or removed at either end but not in the middle is known as _____.

e) If the elements of a data structure are stored sequentially, then it is a _____.

f) Merge sort uses _____ technique.

g) Where is linear searching used?

h) Stack can be implemented using _____ and _____.

versity.onrender.com

i) In C language, malloc() returns _____.

j) What is the number of edges present in a complete graph having n vertices?

k) _____ notation provides a tight lower bound for f(n).

l) Which data structures is used to implement recursion?

Group-B (Short Answer Type Question)

Answer any three of the following:

2. Make a comparison between a linked list and a linear array.

3. A graph contains 21 edges, 3 vertices of degree 4 and all other vertices of degree 2. Find total number of vertices.

4. Write an algorithm for bubble sort technique.

5. Write algorithms for push and pop operations of stack.

6. What are the properties of a B tree? What are the disadvantages of a binary search tree?

Group-C (Long Answer Type Question)

Answer any three of the following:

7. (a) Explain what is ideal Sorting algorithm?

(b) Classify Sorting Algorithms based on some parameter like stability, number of swaps required, number of comparisons, complexity, types, etc.

8. (a) Write merge sort algorithm.

(b) Implement merge sort in C code.

9. (a) Define hashing.

(b) What are the different types of hashing techniques available?

(c) What are the advantages of hashing?

10. (a) Explain with a suitable example the principle of operation of Quick Sort algorithm.

(b) In which cases, Quick Sort becomes a 'Slow Sort'? State its remedy in those cases.

versity.onrender.com

(c) Compare the performance and operation of Bubble Sort and Selection Sort.

11. (a) Define recursion? Explain the usage of stack in recursive algorithm implementation?

(b) What are the types of Queues? Explain each.